



Modelling with Differential Equations

· Mark Scheme

Core Pure 2 Chapter 8: Modelling with Differential Equations. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Forming differential equations

- 1 $\frac{dP}{dt} = kP$ B1; with $k > 0$ since P increases B1.
- 2 (a) $\frac{dN}{dt} = kN(1000 - N)$ B1, with $k > 0$ B1. (b) $N(1000 - N)$ is greatest when $N = 500$ B1.
- 3 Salt in: $0.2 \times 4 = 0.8$ kg per minute M1; concentration $\frac{x}{100}$, salt out $4 \times \frac{x}{100}$ M1; $\frac{dx}{dt} = 0.8 - \frac{x}{25}$ A1.
- 4 (a) $\frac{dV}{dr} = -k(4\pi r^2)$ and $\frac{dV}{dr} = 4\pi r^2 \frac{dr}{dt}$ M1; so $\frac{dr}{dt} = -k$, a constant A1. (b) $r = 6 - kt$, $4 = 6 - 2k \Rightarrow k = 1$ M1; $r = 0$ at $t = 6$, so 4 more hours A1.
- 5(a) Volume $V = 50 + t$ B1; in: $3 \times 0.5 = 1.5$ kg per minute M1; out: $2 \times \frac{x}{50+t}$, giving the result A1.
- 5(b) $\frac{dx}{dt} + \frac{2}{50+t}x = 1.5$, IF $= e^{2\ln(50+t)} = (50+t)^2$ M1 A1; $x(50+t)^2 = \int 1.5(50+t)^2 dt = 0.5(50+t)^3 + c$ M1; $t = 0$, $x = 0$: $c = -0.5 \times 50^3 = -62500$ M1; $x = 0.5(50+t) - \frac{62500}{(50+t)^2}$ A1.
- 5(c) Full when $50 + t = 150$, $t = 100$ M1; $x = 75 - \frac{62500}{22500} = 72.2$ kg A1.

Section B – First-order models

- 6 $P = 2000e^{0.03t}$ M1; $P(10) = 2000e^{0.3} = 2700$ A1.
- 7 $\int \frac{1}{\theta - 20} d\theta = -\int k dt$ M1; $\ln(\theta - 20) = -kt + c$, so $\theta - 20 = Ae^{-kt}$ M1; $t = 0$, $\theta = 90$ gives $A = 70$ A1.
- 8 $40 = 70e^{-5k} \Rightarrow k = \frac{1}{5} \ln \frac{7}{4}$ M1; $k = 0.112$ A1; $20 = 70e^{-kt} \Rightarrow t = \frac{\ln 3.5}{k} = 11.2$ minutes A1.
- 9 $\int \frac{1}{P-200} dP = \int 0.2 dt \Rightarrow P - 200 = Ae^{0.2t}$ M1; $A = -50$: $P = 200 - 50e^{0.2t}$ A1; $P = 0$ when $e^{0.2t} = 4$ M1; $t = 5 \ln 4 = 6.93$ years A1.
- 10(a) $\int \frac{2000}{P(1000-P)} dP = \int dt$ M1; $\frac{2000}{P(1000-P)} = \frac{2}{P} + \frac{2}{1000-P}$ M1; $2 \ln P - 2 \ln(1000-P) = t + c$ A1; $t = 0$, $P = 100$: $c = 2 \ln \frac{1}{9}$, so $\frac{P}{1000-P} = \frac{1}{9}e^{\frac{1}{2}t}$ M1; rearranging, $P = \frac{1000}{1 + 9e^{-\frac{1}{2}t}}$ A1.
- 10(b) $1 + 9e^{-\frac{1}{2}t} = 2 \Rightarrow e^{-\frac{1}{2}t} = \frac{1}{9}$ M1; $t = 2 \ln 9 = 4.39$ years A1.
- 10(c) $P \rightarrow 1000$ birds B1; e.g. P is treated as continuous; no random events such as disease or bad weather; food supply assumed constant B1.

Section C – Second-order models

- 11 $x = A \cos 3t + B \sin 3t$ M1; $\dot{x}(0) = 0 \Rightarrow B = 0$, so $x = 4 \cos 3t$ A1.
- 12 $\omega = \frac{2\pi}{4} = \frac{\pi}{2}$ B1; max speed $a\omega = 0.5 \times \frac{\pi}{2} = 0.785 \text{ ms}^{-1}$ B1; max acceleration $a\omega^2 = 1.23 \text{ ms}^{-2}$ B1.
- 13 Auxiliary equation $m^2 + km + 16 = 0$, compare k^2 with 64: $k = 10$: $k^2 > 64$, real distinct roots, heavy damping B1; $k = 8$: $k^2 = 64$, repeated root, critical damping B1; $k = 2$: $k^2 < 64$, complex roots, light damping (oscillates) B1.
- 14 $m^2 + 6m + 25 = 0 \Rightarrow m = -3 \pm 4i$ M1; $x = e^{-3t}(A \cos 4t + B \sin 4t)$ A1; $A = 2$; $\dot{x}(0) = -3A + 4B = 0$ M1; $x = e^{-3t}(2 \cos 4t + \frac{3}{2} \sin 4t)$ A1.



15(a) $\frac{d^2x}{dt^2} = 0.2\frac{dx}{dt} - 0.2\frac{dy}{dt}$ *M1*; $= 0.2\frac{dx}{dt} - 0.2(0.1x)$ *M1*; $\frac{d^2x}{dt^2} - 0.2\frac{dx}{dt} + 0.02x = 0$ *A1*.

15(b) $m^2 - 0.2m + 0.02 = 0 \Rightarrow m = 0.1 \pm 0.1i$ *M1*; $x = e^{0.1t}(A \cos 0.1t + B \sin 0.1t)$ *A1*.

15(c) $x(0) = 100 \Rightarrow A = 100$ *B1*; $\dot{x}(0) = 0.2(100) - 0.2(20) = 16$ and $\dot{x}(0) = 0.1A + 0.1B$ *M1*;
 $B = 60$: $x = e^{0.1t}(100 \cos 0.1t + 60 \sin 0.1t)$ *A1*.

15(d) $\tan 0.1t = -\frac{5}{3} \Rightarrow 0.1t = \pi - \arctan \frac{5}{3}$, $t = 21.1$ years *M1 A1*. (After this the model gives $x < 0$, so it is no longer valid.)

Roar-level challenge

16 CF $x = A \cos 2t + B \sin 2t$ *B1*; $\sin 2t$ is in the CF, so try $x = t(p \cos 2t + q \sin 2t)$ *M1*; substituting gives $-4p \sin 2t + 4q \cos 2t = 3 \sin 2t$, so $p = -\frac{3}{4}$, $q = 0$ *A1*; $A = 0$, $2B - \frac{3}{4} = 0$: $x = \frac{3}{8} \sin 2t - \frac{3}{4}t \cos 2t$ *A1*; the amplitude grows without limit (resonance), so the model is unrealistic for large t (the spring would break or exceed its elastic limit) *B1*.